



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
YEAR 11 MATHEMATICS EXTENTION
HALF YEARLY EXAM

Mathematics Continuers

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 81

- Attempt questions 1-8
- Start each new section in a separate answer booklet

Examiner: *D.McQuillan*

Total marks – 81

All questions are NOT of equal value

Answer each SECTION in a SEPARATE writing booklet

Section A

Question 1 (10 Marks)	Marks
(a) Evaluate $8^{2.1}$ to 4 significant figures.	2
(b) Write $0.\dot{1}\dot{3}$ as a simplified fraction.	1
(c) Simplify	
(i) $3x - (4 - x)$	1
(ii) $\frac{x+1}{3} + \frac{2x}{5}$	1
(d) Convert 270° to radians in exact form.	1
(e) Factorise	
(i) $x^2 - 9$	1
(ii) $64 + x^3$	1
(f) Given that	
$f(x) = \begin{cases} 6 - x^2 & \text{if } x \geq 0 \\ x & \text{if } x < 0 \end{cases},$	
evaluate	
(i) $f(-2)$	1
(ii) $f(0)$	1

Question 2 (11 Marks)

- (a) Solve $|2x + 6| = 10$. 2
- (b) Simplify $\frac{1}{\sec^2 \theta} + \frac{1}{\operatorname{cosec}^2 \theta}$. 2
- (c) (i) Solve the inequation $|x + 3| < 2$. 2
- (ii) Hence graph the solution on a number line. 1
- (d) Find equation of the circle with centre $(-4, 6)$ and radius $\sqrt{5}$. 2
- (e) Find the exact value of $\tan \frac{3\pi}{4}$. 2

Question 3 (9 Marks)

- (a) (i) Write down the expansion for $\sin(A + B)$. 1
- (ii) Hence find the exact value of $\sin 75^\circ$. 2
- (b) If α and β are the roots of $2x^2 + 3x + 4 = 0$, find the value of:
- (i) $\alpha\beta$ 1
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
- (c) Sketch the intersection of the regions $y > x^2 - 1$ and $y \leq x$. 3

Question 4 (11 Marks)

- (a) State the domain and range of $g(x) = \sqrt{x+4}$ 2
- (b) Show that $f(x) = x^3 + 3x$ is an odd function. 2
- (c) Find correct to the nearest minute the acute angle between $y = 6x - 7$ and $y = x + 3$. 3
- (d) Find the values of A and B if $2(x-1)^2 \equiv A(x^2 + 1) + Bx$. 2
- (e) For $A(5, 1)$ and $B(-3, 7)$ find the coordinates of the point that divides the interval AB internally in the ratio $3:1$. 2

End of Section A

START A NEW WRITING BOOKLET

Section B

Question 5 (9 Marks)

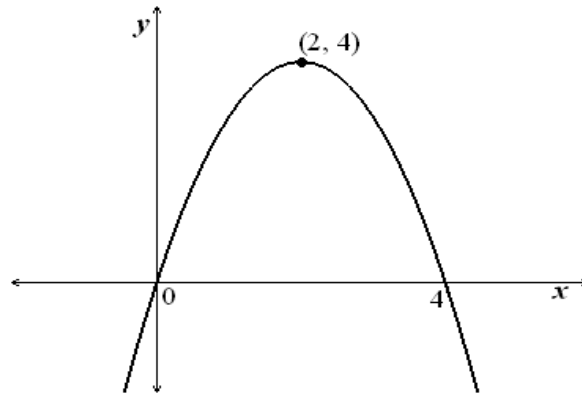
Marks

- (a) The graph of the equation $x^2 - 6x + y^2 + 10y + 9 = 0$ is a circle.
- (i) Explain why the graph of the equation does **not** represent a function. 1
- (ii) Find the radius and the coordinates of the centre of the circle. 2
- (b) Find the equation of a parabola with vertex $(0, -2)$ and focus $(0, 1)$. 2
- (c) Solve the inequality $\frac{x-3}{x} < 0$. 2
- (d) Solve the equation $x^4 - 10x^2 + 9 = 0$. 2

Question 6 (9 Marks)

(a) Sketch the graph of $y = |x + 3|$. 2

(b) Find the equation of the following parabolic graph. 2



(c) Find the values of p for which the equation $x^2 + px + 16 = 0$, has real roots. 2

(d) If $\tan \theta = \sqrt{3}$ and $0 < \theta < \frac{\pi}{2}$ evaluate $\cos 2\theta$ in exact form. 3

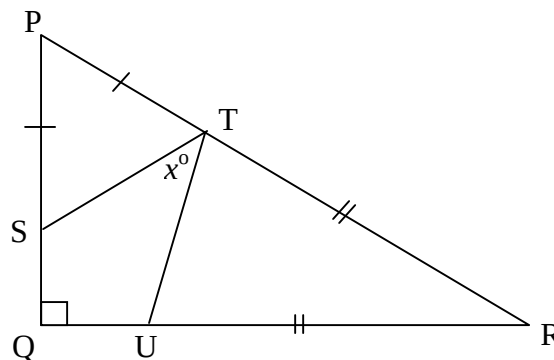
Question 7 (11 Marks)

- (a) Let P be the point $(-2, 3)$ and l be the line $4x - 3y = 8$.
Find the shortest distance from P to l . 2
- (b) Find the equation of the straight line which passes through
the point of intersection of the lines $2x - y - 3 = 0$ and
 $y - 3x - 2 = 0$ and the point $(3, -1)$. 3
- (c) From a point B which is 80 metres due east of a flagpole, F,
the angle of elevation of its top is 15 degrees. Point A is 60
metres due south of the flagpole.
- (i) Draw a neat diagram of the above. 1
- (ii) Find the angle of elevation from point A to the
top of the flag pole. 2
- (d) Solve the following simultaneous equations, 3

$$\begin{aligned}2x + y - z &= -3 \\ -x + 3y - 2z &= 1 \\ x - y + 5z &= 12\end{aligned}$$

Question 8 (11 Marks)

- (a) The quadratic equation $x^2 + Lx + M = 0$ has one root which is twice the other. Prove that
- (i) $2L^2 = 9M$ 3
- (ii) The roots are rational whenever L is rational. 2
- (b) Find the equation of the locus of the point $P(x, y)$ such that it is equidistant from $(3, 1)$ and $y = -1$. 3
- (c) The triangle PQR is right angled at Q and triangles PST and RTU are isosceles as shown. If $\angle STU = x^\circ$ find the value of x . No reasons need to be given. 3



End of Section B

End of Paper

2006 Mathematics Continuers: **Solutions Part A**

Question 1 (10 Marks)

- (a) Evaluate $8^{2.1}$ to 4 significant figures,

2

Solution: By calculator, $78.79324245 \approx 78.79$ to 4 sig. fig.

- (b) Write $0.\dot{1}\dot{3}$ as a simplified fraction.

1

Solution: Let $x = 0.\dot{1}\dot{3}$,
 $100x = 13.\dot{1}\dot{3}$,
 $99x = 13$,
 $\therefore x = \frac{13}{99}$.

- (c) Simplify

- (i) $3x - (4 - x)$

1

Solution: $3x - 4 + x = 4x - 4$

- (ii) $\frac{x+1}{3} + \frac{2x}{5}$

1

Solution: $\frac{5x+5+6x}{15} = \frac{11x+5}{15}$

- (d) Convert 270° to radians in exact form.

1

Solution: $270^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{2}$

- (e) Factorise

- (i) $x^2 - 9$

1

Solution: $(x+3)(x-3)$

- (ii) $64 + x^3$

1

Solution: $4^3 + x^3 = (4+x)(16-4x+x^2)$

(f) Given that

$$f(x) = \begin{cases} 6 - x^2 & \text{if } x \geq 0, \\ |x| & \text{if } x < 0, \end{cases}$$

evaluate

(i) $f(-2)$

1

Solution: $|-2| = 2$

(ii) $f(0)$

1

Solution: $6 - 0^2 = 6$

Question 2 (11 Marks)

(a) Solve $|2x + 6| = 10$.

2

Solution: $2x + 6 = 10$, or $-2x - 6 = 10$,
 $2x = 4$, $-2x = 16$,
 $\therefore x = 2$. $x = -8$.

(b) Simplify $\frac{1}{\sec^2 \theta} + \frac{1}{\operatorname{cosec}^2 \theta}$

2

Solution: $\cos^2 \theta + \sin^2 \theta = 1$

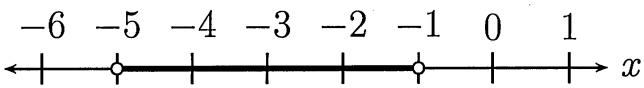
(c) (i) Solve the inequation $|x + 3| < 2$.

2

Solution: $-2 < x + 3 < 2$,
 $-5 < x < -1$.

(ii) Hence graph the solution on a number line.

1

Solution: 

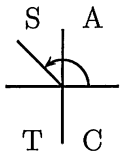
(d) Find the equation of the circle with centre $(-4, 6)$ and radius $\sqrt{5}$.

2

Solution: $(x + 4)^2 + (y - 6)^2 = 5$

- (e) Find the exact value of $\tan \frac{3\pi}{4}$.

2

Solution:  $-\tan \frac{\pi}{4} = -1$

Question 3 (9 Marks)

- (a) (i) Write down the expansion for $\sin(A + B)$.

1

Solution: $\sin A \cos B + \cos A \sin B$

- (ii) Hence find the exact value of $\sin 75^\circ$.

2

Solution: $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2},$
 $= \frac{1 + \sqrt{3}}{2\sqrt{2}} \text{ or } \frac{\sqrt{2} + \sqrt{6}}{4}.$

- (b) If α and β are the roots of $2x^2 + 3x + 4 = 0$, find the value of

- (i) $\alpha\beta$

1

Solution: $\frac{4}{2} = 2$

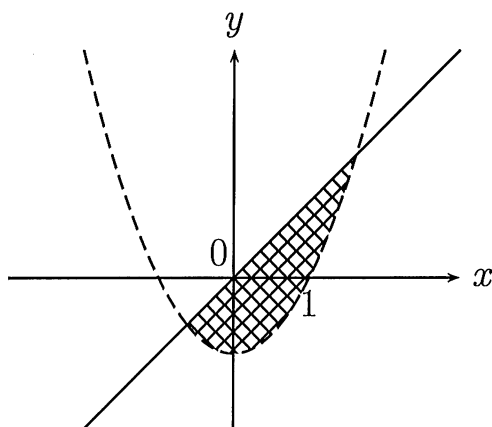
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$

2

Solution: $\frac{\alpha + \beta}{\alpha\beta} = \frac{-3}{2} \times \frac{1}{2},$
 $= -\frac{3}{4}.$

- (c) Sketch the intersection of the regions $y > x^2 - 1$ and $y \leq x$. 3

Solution:



Question 4 (11 Marks)

- (a) State the domain and range of $g(x) = \sqrt{x+4}$. 2

Solution: Domain: $x \geq -4$,
Range: $g(x) \geq 0$.

- (b) Show that $f(x) = x^3 + 3x$ is an odd function. 2

Solution: $f(-x) = (-x)^3 + 3(-x)$,
 $= -x^3 - 3x$,
 $= -(x^3 + 3x)$,
 $= -f(x)$.
 $\therefore f(x)$ is odd.

- (c) Find correct to the nearest minute the angle between $y = 6x - 7$ and $y = x + 3$. 3

Solution: $m_1 = 6$, $m_2 = 1$.
 $\tan \alpha = \left| \frac{6 - 1}{1 + 6 \times 1} \right|$,
 $= \frac{5}{7}$.
 $\therefore \alpha = 35^\circ 32'$.

(d) Find the values of A and B if $2(x - 1)^2 \equiv A(x^2 + 1) + Bx$.

2

Solution: Let $x = 0$, $2 = A$.
Let $x = 1$, $0 = 4 + B$,
 $B = -4$.

(e) For $A(5, 1)$ and $B(-3, 7)$ find the coördinates of the point that divides the interval AB internally in the ratio $3 : 1$.

2

Solution: $\left(\frac{3 \times (-3) + 1 \times 5}{3 + 1}, \frac{3 \times 7 + 1 \times 1}{4} \right) = (-1, 5\frac{1}{2})$

711 CONTINUOUS

SECTION B

5 (a) $x^2 - 6x + y^2 + 10y + 9 = 0$

$$(x-3)^2 + (y+5)^2 = -9 + 9 + 25 \\ = 25.$$

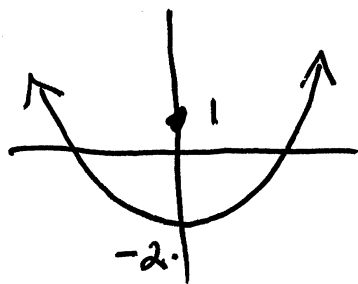
(i) This circle doesn't pass the vertical line test for a function. ✓

OR. There are points (ie two) where the x -value is the same.

OR. It is not a 1-1 correspondence & mapping

(ii) Circle $(3, -5)$ radius 5. ✓✓

(b)



NB $a = 3$.

$$\therefore (x-0)^2 = 4 \times 3 (y+2)$$

$$\boxed{x^2 = 12(y+2)} \quad \checkmark \checkmark$$

(c) $\frac{x-3}{x} < 0$

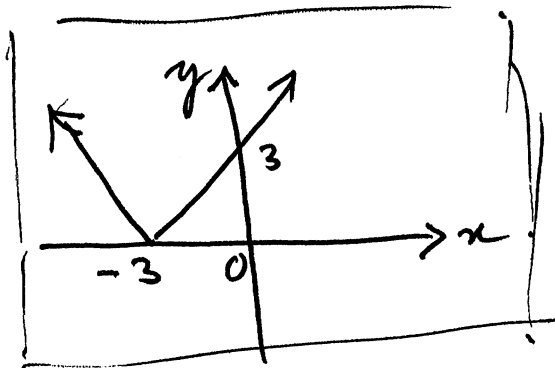
$x(x-3) < 0$ (ie multiply both sides by x^2)

$$\therefore \boxed{0 < x < 3} \quad \checkmark \checkmark$$

(d) $x^4 - 10x^2 + 9 = 0$ let $u = x^2$ $u = 9, 1$ ✓✓✓
 $u^2 - 10u + 9 = 0$ ie $x^2 = 9, 1$
 $(u-9)(u-1) = 0$ $\boxed{x = \pm 3, \pm 1}$

6

(a)



$$(b) \quad y = -a(x-0)(x-4)$$

$$\therefore y = -ax(x-4)$$

now $(2, +4)$ lies on it.

$$\therefore +4 = -2a \cdot -2.$$

$$\therefore \underline{a = 1}$$

$$\therefore \boxed{y = -x(x-4)}$$

$$(c) \quad \text{For real roots } \Delta \geq 0$$

$$\therefore p^2 - 64 \geq 0$$

$$(p-8)(p+8) \geq 0$$

$$\therefore \boxed{p \geq 8, p \leq -8.}$$

$$(d) \quad \tan \theta = \sqrt{3}, \quad 0 < \theta < \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

$$\therefore \boxed{\cos 2\theta = \cos \frac{2\pi}{3} = -\frac{1}{2}.}$$

Q7

(a) $4x - 3y - 8 = 0$ and point $(-2, 3)$

$$\begin{aligned} d &= \left| \frac{4(-2) - 3(3) - 8}{\sqrt{4^2 + (-3)^2}} \right| \\ &= \left| \frac{-8 - 9 - 8}{5} \right| \\ &= \left| -\frac{25}{5} \right| \\ &= 5 \end{aligned} \quad \checkmark \checkmark$$

(b) Find the intersection of $2x - y = 3$ — (1)
 $3x - y = -2$ — (2)

$$(2) - (1)$$

$$x = -5$$

Sub in (1)

$$-10 - y = 3$$

$$\underline{y = -13}$$

$$\therefore (-5, -13)$$

$\therefore$ Equation of line

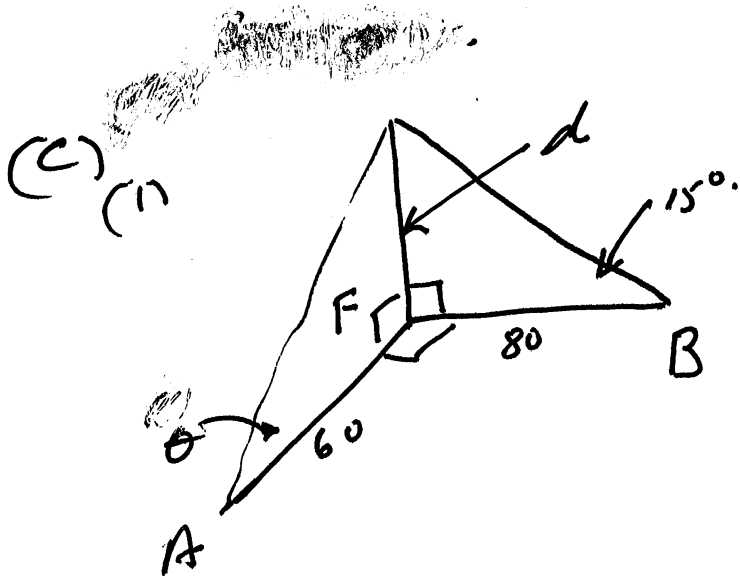
having through $(3, -1)$ and $(-5, -13)$

$$\begin{aligned} \frac{y+1}{x-3} &= \frac{-13+1}{-5-3} = \frac{-12}{-8} \\ &= \frac{3}{2} \end{aligned}$$

$$\therefore 2y+2 = 3x-9$$

$$\underline{3x - 2y - 11 = 0}$$

$\checkmark \checkmark \checkmark$



(11) $\frac{d}{80} = \tan 15^\circ$
 $\therefore d = 80 \tan 15^\circ$

now $\tan \theta = \frac{d}{60}$
 $= \frac{80 \tan 15^\circ}{60}$
 $= 0.3573$

$\therefore \theta = 19^\circ 40' \text{ (OR } 20^\circ)$

(d) $2x + y - z = -3$ — (1)

$-x + 3y - 2z = 1$ — (2)

$x - y + 5z = 12$ — (3)

(1) + (3)

$3x + 4z = 9$ — (4)

(3) $\times 3$

$3x - 3y + 15z = 36$ (3a)

(3a) + (2)

$2x + 13z = 37$ (5)

(4) $\times 2$

$6x + 8z = 18$ — (4a)

(5) $\times -3$

$-6x - 39z = -111$ (5a)

(4a) + (5a)

$-31z = -93$

$z = 3$

Sub in (4)

$3x + 12 = 9$

$3x = -3$

$x = -1$

Sub in (1)

$-2 + y - 3 = -3$

$y = 2$

$\therefore (-1, 2, 3)$

Q8

(a) $x^2 + Lx + M = 0$ has roots $\alpha, 2\alpha$.

Now $\alpha + 2\alpha = -L$ & $\alpha \times 2\alpha = M$

(i)

$\therefore 3\alpha = -L$ — (1)

$\alpha = -\frac{L}{3}$

$2\alpha^2 = M$ — (2)

From (1) & (2)

$2 \times \left(-\frac{L}{3}\right)^2 = M$

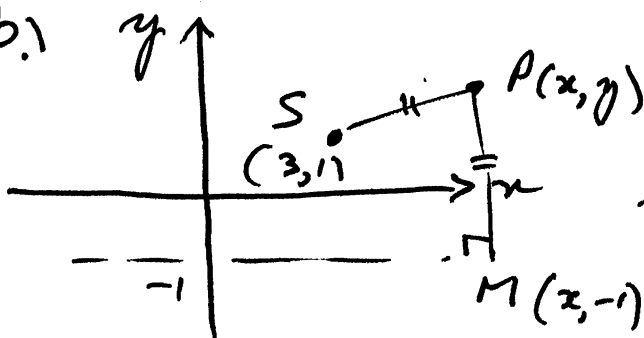
$\frac{2L^2}{9} = M$

✓✓✓

$\boxed{2\alpha^2 = 9M}$

(ii) If L is rational then $-\frac{L}{3} = \alpha$ is rational ✓
 & hence $2\alpha = -\frac{2L}{3}$ is also rational ✓

(b)



Now $SP = PM$

i.e. $\sqrt{(x-3)^2 + (y-1)^2} = y+1$

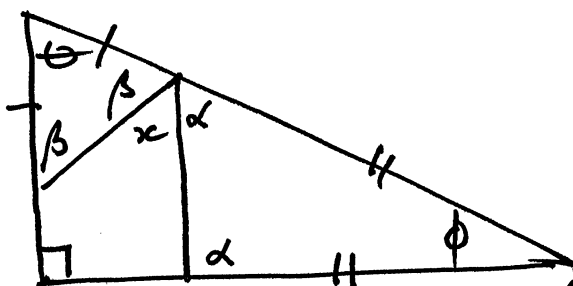
$(x-3)^2 + (y-1)^2 = (y+1)^2$

$x^2 - 6x + 9 + y^2 - 2y + 1 = y^2 + 2y + 1$

$\boxed{(x-3)^2 = 4y}$

✓✓✓

(c)



Now

$\alpha + \beta + x = 180^\circ$ — (1)

$\theta + \phi = 90^\circ$

i.e. $(180 - 2\beta) + (180 - 2\alpha) = 90^\circ$ ✓✓

i.e. $2\alpha + 2\beta = 270^\circ$

$\alpha + \beta = 135^\circ$ — (2)

From (1) & (2) $\boxed{x = 45^\circ}$